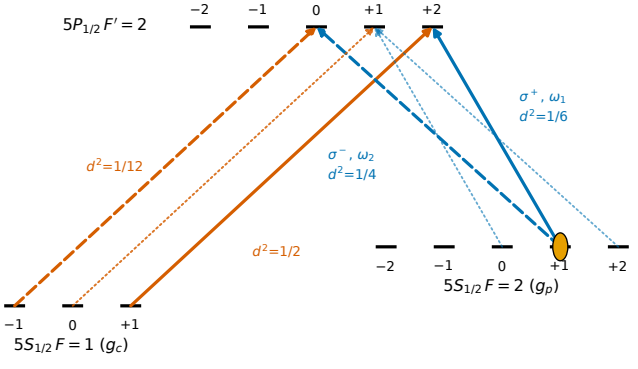
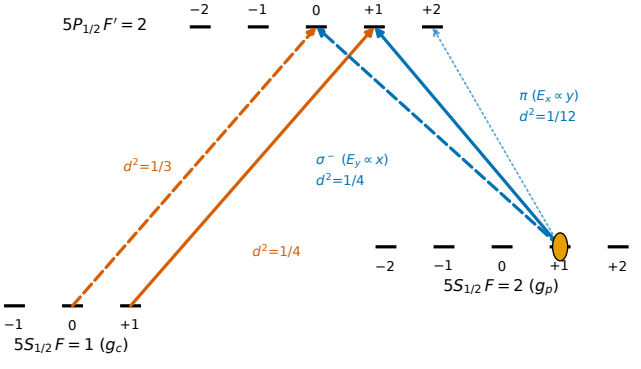


Raman shifts ($g_1 = -\frac{1}{2}$, $g_2 = +\frac{1}{2}$): $|2,1\rangle \to |1,1\rangle$: $-\mu_B B$; $|2,1\rangle \to |1,-1\rangle$: 0.
Dotted: spectators $|2,2\rangle$ (σ^- only) and $|2,0\rangle$ (σ^+ only), both via $|1,0\rangle$.



Raman shifts: $|2,1\rangle \to |1,1\rangle$: $-\mu_B B$; $|2,1\rangle \to |1,0\rangle$: $-\frac{1}{2}\mu_B B$.
Dotted: σ^+ to $|2',2\rangle$ has no partner in $F = 1$ (π coupling).